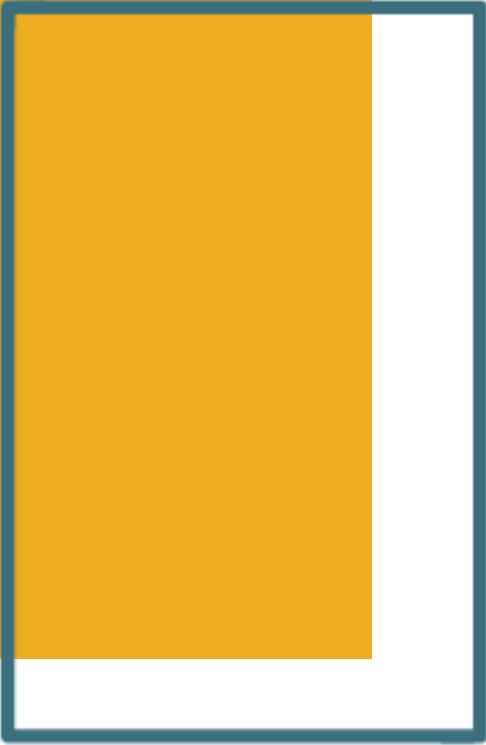




04

Image Processing 2



Blurring and Smoothing

- In simple terms, blurring happens when your camera takes a picture out of focus. Sharper regions in the image lose their detail.
- The goal here is to use a low-pass filter to reduce the amount of noise and detail in an image.
- Practically, this means that each pixel in the image is mixed in with its surrounding pixel intensities. This “mixture” of pixels in a neighborhood becomes our blurred pixel.
- Smoothing and blurring is one of the most common pre-processing steps in computer vision and image processing.

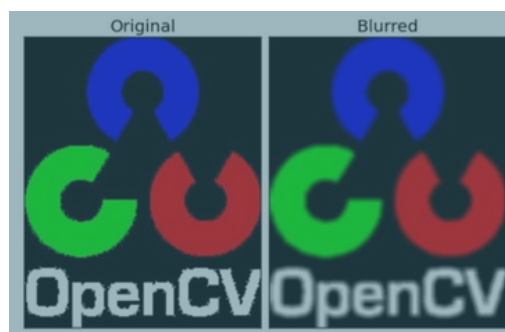
- Therefore removing noises and controlling the intensity of the pixel values can help the model to focus on the general details and get higher accuracy.
- By smoothing an image prior to applying more techniques, we are able to reduce the amount of high frequency content/detail, such as noise and edges (i.e. the “detail” of an image).
- This allows us to focus on the larger structural objects in the image.

There are several techniques used to achieve blurring effects but we’re going to talk about the four major ones used in OpenCV: Averaging blurring, Gaussian blurring, Median blurring, and Bilateral filtering.

Averaging blurring



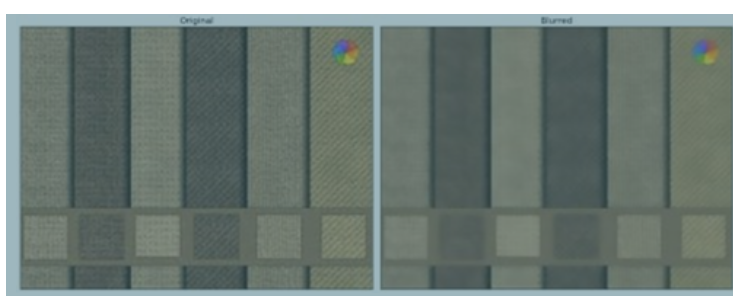
Gaussian blurring



Bilateral filtering



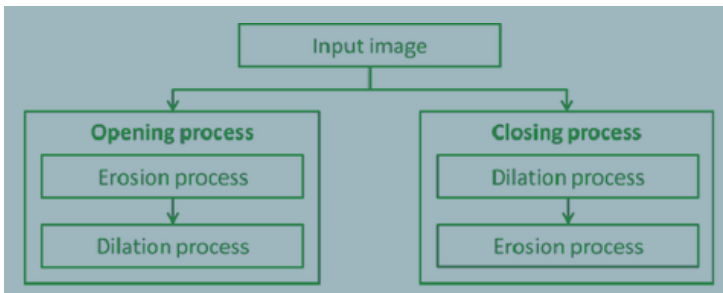
Median blurring



Morphological Operator

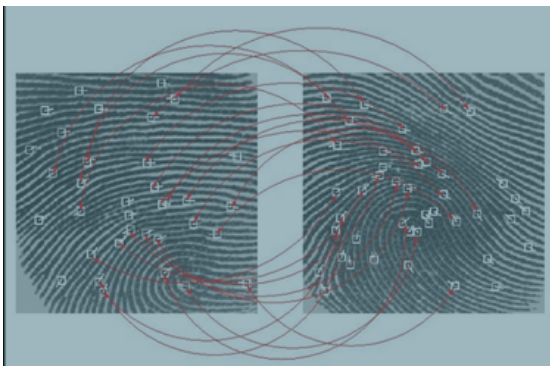
- Morphological operators are sets of Kernels that can achieve a variety of effects, such as reducing noise.
- Certain operators are very good at reducing black points on a white background (vice versa)
- There are some operator in morphological techniques: Erosions & dilation, opening, closing
- Dilation adds pixels to the boundaries of objects in an image, while erosion removes pixels on object boundaries.
- The number of pixels added or removed from the objects in an image depends on the size and shape of the structuring element used to process the image.

- Opening is a process in which first erosion operation is performed and then dilation operation is performed.
- Closing is a process in which first dilation operation is performed and then erosion operation is performed.

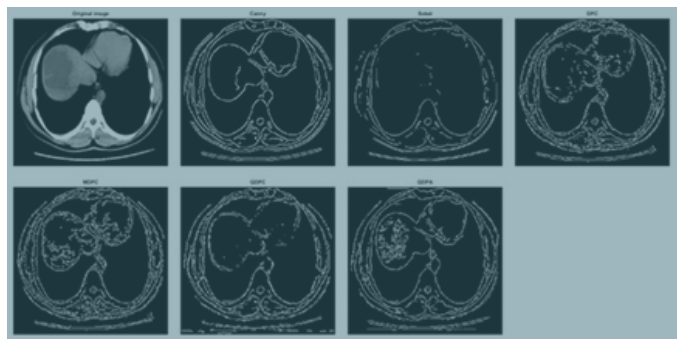


Gradients

- An image gradient is a directional change in the intensity or color in an image. The gradient of the image is one of the fundamental building blocks in image processing.
- Usually, image gradient is used for edge detection.
- Edge detection is pervasive in several applications such as finger-print matching, medical diagnosis and license plate detection.
- These applications basically highlights the areas where image intensity changes drastically and ignore everything else.



Finger printing



Medical diagnosis

Picture Source

- unsplash.com
- pexels.com
- opencv.org